Margeaux Corrigan

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EDUCATION

Brown University, *Sc. B. Electrical Engineering*, 4.0/4.0 GPA — Providence, RI | **Expected Graduation May 2029**

Relevant Courses: Ordinary Differential Equations, Functional Programming, Statics & Dynamics, Data Structures & Algorithms

TECHNICAL SKILLS

Programming: C/C++, Python (NumPy, Pandas, Matplotlib), Ada, MATLAB, Java, OCaml, Racket

Embedded Systems: Bare-metal firmware (RP2040, STM32), register-level programming, I2C/SPI, hardware bring-up

Hardware: PCB design, soldering, CAD & CAM(SolidWorks, Fusion 360, Onshape), FEA, 3D printing, sheet metal fabrication

RESEARCH

Interactive 3D Vision & Learning Lab (IVL), *Incoming Paid Researcher* — Providence, RI | Summer 2026

Advisor: Srinath Sridhar, Assistant Professor | Brown University

- Selected to investigate multi-camera capture and calibration systems using C++ and embedded hardware

INDUSTRY EXPERIENCE

AdaCore, *Technical Mentorship* — Providence, RI | April 2026 - Present

Mentor: Oliver Henley, UX Engineer | AdaCore

- Developed bare-metal Ada firmware for STM32G431 using direct register-level programming (RM0440/SVD), without HAL
- Parsed ARM SVD/XML files to generate register mappings, enabling faster development of STM32 firmware in Ada
- Configured STM32 GPIO peripherals directly from reference manual without HAL, validating functionality on hardware

Persistence Labs, *Strategy Intern* — Charleston, SC | May 2024 – October 2025

- Facilitated beta testing with high school control groups, collected feedback across 10+ iterations to identify usability issues
- Synthesized testing results into reports and presented findings to the founder/CEO, guiding UI and product design decisions
- Communicated user feedback to engineering team, influencing feature prioritization and UI design decisions

MIG3 Inc., *Part-Time Technical Assistant* — Irvine, CA | May 2022 – August 2025

- Provided technical support for office software and built Excel-based data tools, improving internal workflow efficiency

LEADERSHIP & ACTIVITIES

Brown Rocketry Club, *Avionics Team Lead* — Providence, RI | September 2025 – Present

- Contributed to Brown Rocketry's first-ever IREC acceptance as sole avionics developer on the competing payload
- Designed IREC avionics electrical architecture and wiring diagrams for power distribution, sensor interfaces, and flight systems
- Performed FEA on avionics bay components and developed CAM-based manufacturing plans in Fusion 360
- Secured annual industry sponsorship by leading partner outreach and presenting club technical work to external stakeholders

Brown Space Engineering, *Antenna Team Member* — Providence, RI | March 2026 – Present

- Designing communications antennas (Nitinol dipole, copper patch), including specification and integration constraints
- Establishing feed architecture, impedance matching strategy, and EMC constraints on adjacent and connected hardware

AI & Robotics Ethics Society, *Research Team Member* — Providence, RI | September 2025 – Present

- Designing a survey and assessment to evaluate the effectiveness of Brown's Socially Responsible Computing curriculum
- Collaborating with the Computer Science Department and teaching assistants to collect and analyze student feedback

Project Amplify, *Founder & Director* — Irvine, CA | May 2024 – September 2025

- Founded a summer AI literacy program across 3+ school districts, designed and taught curriculum from scratch
- Taught students to train basic image recognition AI using Python, introducing coding and model evaluation with Pandas/NumPy

ENGINEERING PROJECTS

Open Source Bare-Metal Driver Stack, *Independent Project* — C | April 2026 – Present

- Developing device-portable bare-metal drivers for GPIO, I2C, UART, and SysTick timing with direct register-level control
- Building device drivers (LSM9DS1, BMP390) and a hardware abstraction layer separating registers from portable logic

Rocket Payload Sensor System, *Sole Designer* — RP2040, Ada | February 2026 – April 2026

- Designed and implemented bare-metal Ada firmware for an RP2040-based multi-sensor rocket payload, mentored by AdaCore
- Deployed payload on live high-power rocket launch; logged repeatable 12+ G flight data, validating system end to end
- Integrated BMP390 & LSM9DS1 over I2C at 100 Hz in testing; implemented register-level config and data acquisition
- Implemented W25Q128 SPI flash driver for onboard data logging; validated full write/erase/read-back pipeline on hardware
- Debugged hardware issues including I2C addressing errors, SPI clocking, UART data loss, and power sequencing

AWARDS & GRANTS

Brown UTRA Research Award, 2026 | **Brown Design Workshop Grant**, 2026 | **Taco Bell Ambition Accelerator Grant,** 2025